

sends them to a browser and retrieves the result. It does not require any special server to execute tests. Selenium Grid can be used with Web Driver to execute tests on remote systems. At present it is fully implemented and supported in Ruby, Python, C# and Java.

Selenium Grid:

This is a server that allows tests to use Web browsers running on remote systems. In this, one server acts as the hub. The tests contact the hub to obtain access to browsers. The hub consists of lists of servers that provide access to browsers (i.e., Web Driver nodes). It allows the running of tests in parallel on multiple machines, and the management of different versions and configurations of browsers, centrally. This helps to spread the load of testing across several machines by running the tests across different platforms or operating systems.

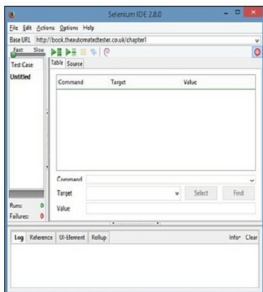


Figure 5: Selenium IDE

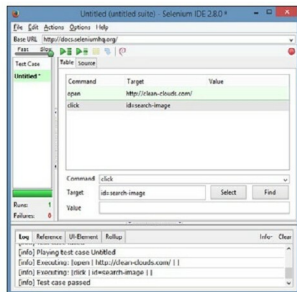


Figure 6: Sample test case execution on Selenium IDE

Installation of Selenium

1. Open a Mozilla Firefox browser.
2. Type the URL <http://seleniumhq.org/download/> in the browser and download the latest version as specified in the screenshot.
3. When the download link is clicked, Firefox will enable a pop-up that asks if you want to allow Mozilla Firefox

A comparison between various automation testing tools

Parameters	Selenium	QTP	Watir	GEB
Licence cost	Open source software	HP licensed software	Open source software	Open source software
Scripting language	Java, C#, Python, Ruby, PHP, Perl, JavaScript	VB Script	Ruby	Groovy
Operating system support	Windows, Linux, Solaris	Windows 8/XP/ Vista	Windows, Linux, Solaris	Windows, Linux, DOS
Browser support	Firefox, IE, Opera, Safari, Chrome	IE, Firefox, Chrome	Firefox, IE, Opera, Safari, Chrome	Firefox, IE, Opera, Safari, Chrome
Object recognition	Selenium IDE, Firebug, Firepath	Using Object Spy	Open Twebst (Web recorder)	Using GEB IDE
Script execution speed	Low	High	Low	Low
Non-browser based app support	No	Yes	No	No
Device support	Supports iOS and Android	iOS, Android, Blackberry, Windows	iOS and Android	iOS
Ease of support	User and professional community support is available	Has dedicated HP support	Limited support from open source community	Limited support from open source community
Script creation time	Time required is much more	Time required is less	Time required is more	Time required is more